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Blockchain-based system for carbon accounting in intermodal transportation

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Abstract: Supply chain logistics significantly contribute to global CO_2 emissions, with transportation as the largest source of emissions. Carbon accounting plays a critical role in monitoring, measuring, and reporting emissions across supply chains, helping to identify key carbon-intensive areas. This is important in intermodal transportation, where switching between different transportation modes optimizes efficiency and cost but complicates emissions tracking and accountability of emissions. Current methods lack transparency, hindering effective monitoring. Blockchain offers a decentralized solution, enabling secure and transparent carbon accounting. This paper proposes a blockchain-based system for carbon accounting in intermodal transportation emissions using Ethereum smart contracts.

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Keywords: Carbon Accounting, Intermodal Transportation, Sustainability, Life Cycle Analysis

1. INTRODUCTION

The transition towards a net-zero carbon economy, in alignment with the goals of the Paris Agreement at both national and global levels, necessitates the rapid decarbonisation of supply chains and industrial systems (Halkos and Nomikos, 2021). Emissions from transportation activities contribute up to 50% of global CO_2 emissions in supply chains. These systems are inherently complex, involving interconnected global supply chains spanning multiple regions (Acquaye et al., 2018). Effective decarbonisation of the intermodal transportation network, which involves multiple supply chain stakeholders across different modes of transportation, must address all three types of carbon emissions: Scope 1 (direct emissions), Scope 2 (indirect emissions from purchased energy), and Scope 3 (other indirect emissions). Traditionally, Scope 1 emissions, which involve direct emissions from supply chain activities such as vehicle emissions, have been the primary focus in climate science and policy (Barrett et al., 2013). However, Scope 2 and Scope 3 emissions, including those from electricity consumption and other indirect activities such as upstream emissions from fuel production, emissions from the production of vehicles and infrastructure, etc.—often overlooked in carbon emissions inventories—also contribute significantly to the overall carbon footprint (Hertwich and Wood, 2018). While research has explored blockchain's potential to track products across supply chains, studies specifically focused on tracking carbon emissions remain limited (Ju et al., 2022). Existing systems also struggle to effectively monitor and account for all Scope 1, 2, and 3 emissions within a single platform, making it challenging to identify key emission sources and develop strategies for emission reduction (Tardivo and Martín, 2023; Sun et al., 2023). Furthermore, most current systems primarily track only their Scope 1 emissions, as tracking Scope 2 and Scope 3 emissions risks double counting. This limitation makes it more difficult to identify carbon hotspots and develop effective emission reduction strategies (da Cruz et al., 2020).

To effectively reduce carbon emissions, it is crucial and timely to adopt a comprehensive approach that provides full visibility and traceability across global intermodal transportation services (Craig et al., 2013; Alves et al., 2024). Addressing these challenges involves both methodological and practical obstacles. Methodologically, the traceability and accounting model must evaluate Scope 1, 2, and 3 emissions, encompassing both direct and indirect emissions across various regions within an integrated framework. Practically, the global scope of supply chains restricts the effectiveness of existing traceability and accounting systems when used in isolation.By leveraging blockchain technology, a carbon accounting system for intermodal transportation can be developed to create a carbon traceability system with tracking capabilities across multiple regions. Blockchain operates as a decentralized, immutable ledger that records transactions securely and transparently. Each carbon emission entry is cryptographically linked to previous records, ensuring data integrity and preventing unauthorized alterations. Smart contracts can automate verification processes, enabling

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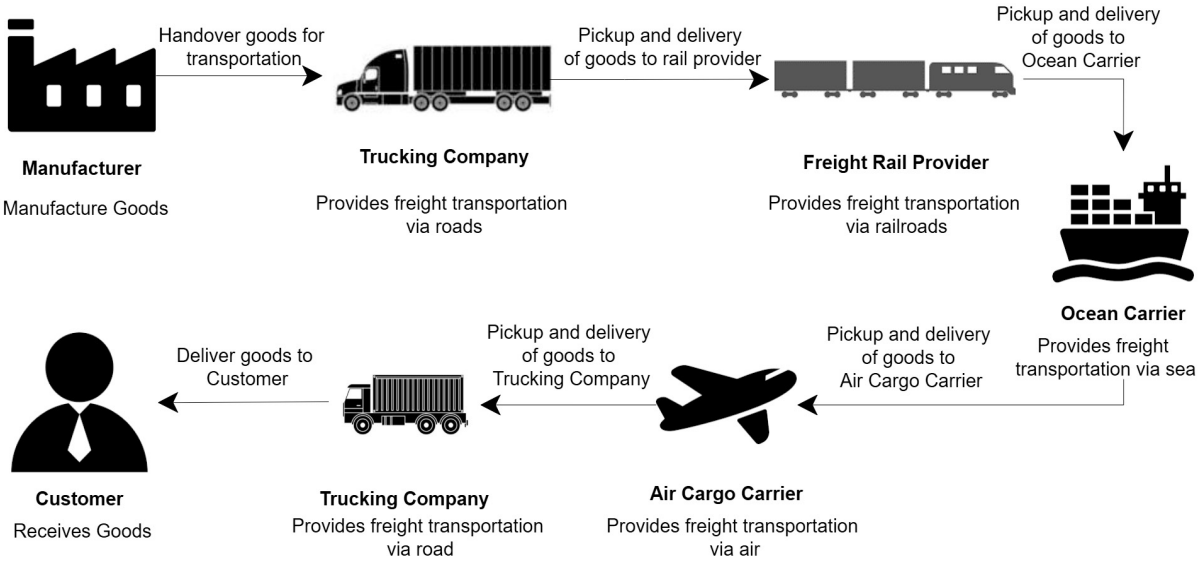

Fig. 1. An Overview of Goods Transportation through Intermodal Transportation

real-time tracking of emissions and streamlining compliance with regulatory frameworks. This enhances accountability among stakeholders and facilitates accurate carbon footprint assessments across the supply chain (Effah et al., 2021; Craig et al., 2013). This system addresses challenges related to transparency, accountability, and methodological issues such as system boundary limitations, verification, and data exchange between intermodal transportation partners. Figure 1 presents an overview of an intermodal transportation case study (Craig et al., 2013; Alves et al., 2024). In this paper, we propose a blockchain-based system for carbon accounting to enhance carbon traceability and accounting in the intermodal supply chain transportation system. The main objectives of this paper are as follows:

- We present a blockchain-based system for effective carbon traceability and accounting in intermodal transportation operations.
- We discuss the role of intermodal supply chain stakeholders and examine how they interact with the blockchain system for carbon traceability and accounting.
- We design and deploy the proposed solution using smart contracts on the Ethereum blockchain (Mohammed et al., 2021; Buterin, 2016).
- We discuss the generalisation of the proposed system so that it can be adapted to other sectors.

We also make the smart contract publicly available on Github [1] . The structure of the paper is as follows: Section 2 presents the literature review, Section 3 outlines the system architecture, Section 4 details the system testing and implementation, and Section 5 presents the conclusion.

2. LITERATURE REVIEW

Due to the rapid increase in CO_2 emissions over the past two decades, the UN Intergovernmental Panel on Climate Change (IPCC) has urged countries to achieve a 50% reduction in greenhouse gas emissions across all sectors by 2030, with a focus on reaching net-zero targets (Intergovernmental Panel on Climate Change (IPCC), 2022). This goal can be realized by accurately tracking and reducing carbon-emitting activities. In the context of intermodal transportation, carbon accounting begins with identifying emission sources, followed by tracking and recording CO_2 emissions across Scope 1, 2, and 3. Establishing emission factors, calculation standards, and management frameworks are crucial components of carbon calculation in this sector (Qian et al., 2024; Mishra et al., 2024). Within complex intermodal supply chains, precise and transparent tracking of carbon emissions is essential for reliable carbon accounting and reporting (Howarth and Winfield, 2023). One of the major challenges in carbon accounting within the transportation industry is the unavailability of reliable data, which complicates effective corporate carbon management. Additionally, there is a lack of suitable concepts and tools for measuring carbon performance throughout the entire intermodal supply chain, from production processes to end-use (Oudani, 2023).

Digital technologies, such as Blockchain, can significantly enhance carbon accounting and reporting in intermodal transportation by ensuring transparency and security. The adoption of Blockchain across sectors like automotive, food, and manufacturing demonstrates its effectiveness (Fareed et al., 2024; Kaack et al., 2018; Thirunavukkarasu et al., 2023; Vilkov and Tian, 2023). Blockchain's characteristics—decentralization, transparency, security, and trust—make it an ideal solution for secure, tamper-proof data storage (Thirunavukkarasu et al., 2023). However, the application of Blockchain in carbon markets has primarily focused on carbon trading, offsets, and traceability, with limited research on comprehensive carbon accounting and full traceability of the carbon footprint (Vilkov and Tian, 2023). Table 1 presents additional state-of-the-art techniques in carbon emissions accounting and blockchain applications across various transportation sectors, along with their features, methodologies, and research gaps, which this paper aims to address.

[1] https://github.com/PratyushKumarPatro/Intermodal-Transportation-Carbon-Accounting/tree/main

Table 1. Summary of literature review and research gaps

Paper	Sector	Features	Methodology	Research Gap
(Koh et al., 2013)	All	CO_2 traceability & accounting	Hybrid life cycle assessment	Centralised system
(Oudani, 2023)	Intermodal	Transportation cost & carbon emissions reduction	Multi-objective optimization model	No blockchain prototype
(Serra et al., 2022)	Maritime and port industry	Real-time Data Sharing and interport connectivity	SWOT Analysis & system framework	Not implemented for carbon accounting and traceability
(Sun et al., 2023)	Maritime and port industry	Green certification	Principal–agent theory	No blockchain-based prototype solution
(Wang et al., 2024)	Aviation industry	Carbon emissions reporting and accounting	Spatial index decomposition	Centralised system
(Mei et al., 2021)	Freight transportation and logistics sector	Decision-making under uncertainty and Revenue optimization	Stochastic optimization model & Empirical analysis	No prototype implementation
(Peng et al., 2024)	Road and rail transportation	Emissions trading system, monitoring, reporting, and verification (MRV) of emissions	Comprehensive literature review	No prototype implementation
(Shao et al., 2023)	Transportation & logistics sector	Carbon emission optimization, Multi-factor emission model	Multi-modal transport carbon emissions calculation model	No prototype solution with system stakeholders
(Lu et al., 2022)	Road transportation	Decentralised carbon emissions trading and auditing	Blockchain-based system framework	Not applied on inter-model transportation systems with multiple stakeholders
(Heiss et al., 2023)	Transportation	Carbon traceability, accounting, and offsetting	Blockchain-based system framework	No generalisation analysis
(Woo et al., 2021)	Building and construction sector	Energy performance measurement, reporting, verification (MRV)	Literature review	No prototype solution
(Qian et al., 2024)	Multimodal transportation & logistics	Trust and information sharing, Transport document collaboration	Blockchain-based framework	No specific to carbon emissions traceability and accounting

In summary, while existing state-of-the-art techniques effectively use blockchain for carbon emissions management within various transportation sectors, there remains a significant research gap in developing a comprehensive approach for end-to-end traceability and accounting of Scope 1, 2, and 3 emissions. This paper addresses this gap by proposing a novel framework that integrates blockchain technology with advanced carbon accounting methods to ensure greater transparency, accuracy, and real-time tracking across the entire supply chain. Our work contributes to advancing carbon management practices by providing a scalable and secure solution for the full scope of emissions, which has been overlooked in previous studies.

3. BLOCKCHAIN-BASED SYSTEM ARCHITECTURE

In this section, we present the proposed blockchain-based system architecture for carbon accounting in intermodal transportation. Figure 2 provides an overview of our Blockchain-based carbon accounting system's architecture. The system includes seven stakeholders, each with specific roles and responsibilities detailed below.

This system enables end-to-end traceability and accounting of the carbon footprint in the context of intermodal transportation. Each stakeholder within the system is uniquely identified by an Ethereum address. The

Table 2. Ethereum address of stakeholders

Stakeholder	Ethereum Address
Regulatory Authority	0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
Trucking Company 1	0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2
Manufacturer	0x4B20993Bc481177ec7E8f571ceCaE8A9e22C02db
Freight Rail Provider	0x78731D3Ca6b7E34aC0F824c42a7cC18A495cabaB
Ocean Carrier	0x617F2E2fD72FD9D5503197092aC168c91465E7f2
Air Cargo Carrier	0x17F6AD8Ef982297579C203069C1DbfFE4348c372
Trucking Company 2	0x5c6B0f7Bf3E7ce046039Bd8FABdfD3f9F5021678
Customer	0x03C6FcED478cBbC9a4FAB34eF9f40767739D1Ff7

Ethereum addresses for all participants in our model are outlined in Table 2. For example, the Ethereum address '0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2' has been assigned to Trucking Company 1.

- Manufacturer: The Manufacturer records Scope 1, 2, and 3 CO_2 emissions generated from the production of goods, including raw material procurement, processing, and energy consumption. These emissions are logged using smart contracts to ensure accuracy and transparency.
- Trucking Company 1 and 2: Trucking companies are responsible for recording Scope 1, 2, and 3 CO_2 emissions generated during their logistics and transportation activities. This includes emissions from fuel consumption, vehicle operations, and associated supply chain processes, updated directly to the blockchain ledger.

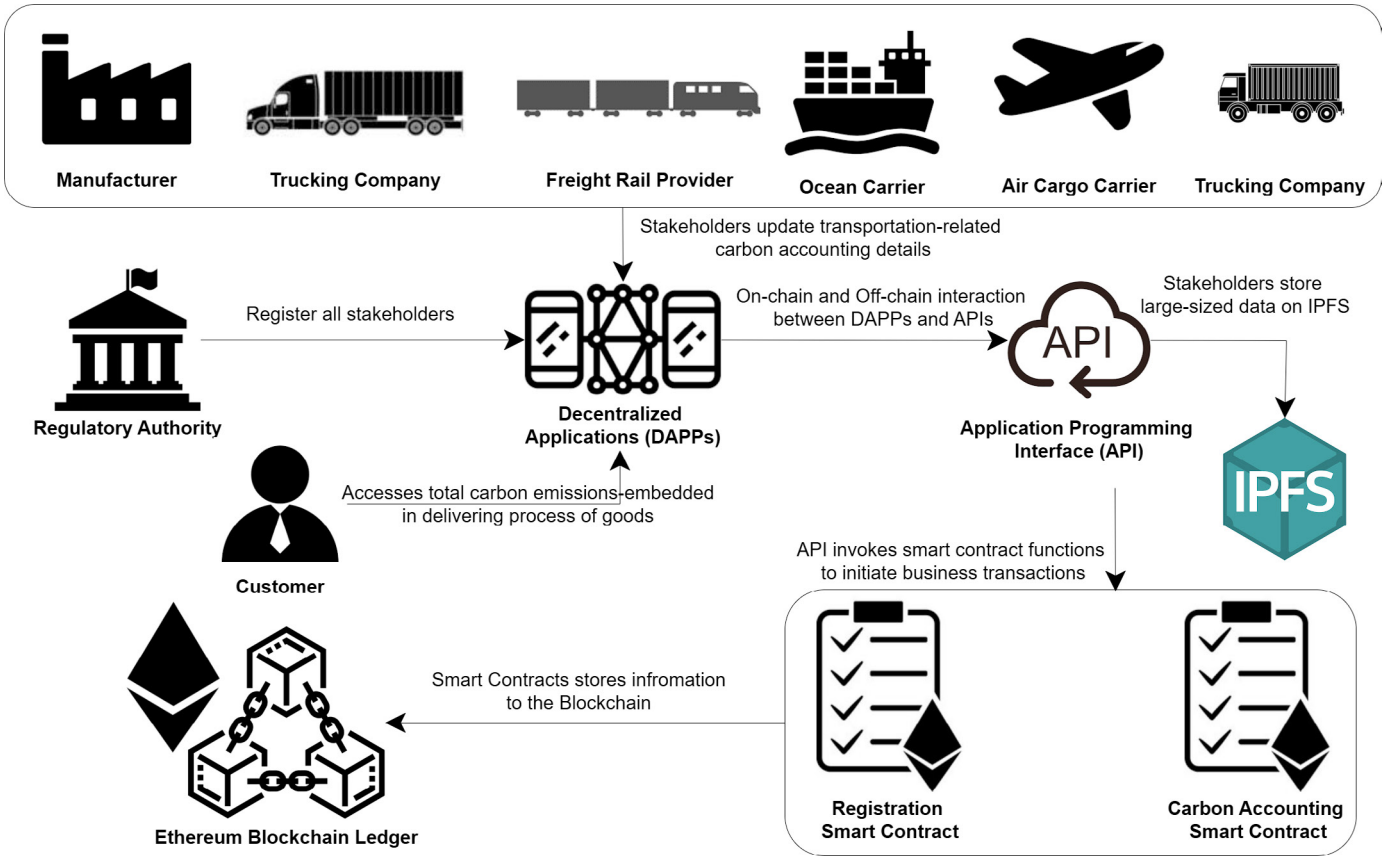

Fig. 2. Blockchain-based system architecture for intermodal transportation carbon accounting

- Freight Rail Provider: The Freight Rail Provider inputs Scope 1, 2, and 3 CO_2 data related to rail logistics and operations, including energy consumption and emissions generated during transportation of goods via rail.
- Ocean Carrier: Ocean carriers record Scope 1, 2, and 3 CO_2 emissions produced during maritime shipping. This includes emissions from vessel operations, port activities, and fuel consumption for long-haul transport across oceans.
- Air Cargo Carrier: Air cargo providers log Scope 1, 2, and 3 CO_2 emissions from air freight services, including emissions related to aircraft fuel usage, cargo handling at airports, and associated supply chain activities..
- Customer: Customers contribute to the carbon footprint through the usage and maintenance of products. The customer accesses the related carbon footprints generated from the transportation network using DAPPs.
- Decentralized Applications (DApps): Decentralized applications streamline the interaction between stakeholders and the Blockchain. They provide an accessible interface for recording, managing, and retrieving carbon emission data.
- Application Programming Interfaces (APIs): APIs enable automated data exchange between Blockchain and external systems or DApps, ensuring seamless interoperability and efficient carbon data management across the intermodal network.
- Smart Contracts: Each stakeholder leverages smart contracts to record Scope 1, 2, and 3 CO_2 emissions from their respective operations. Registration smart contracts ensure all participants are onboarded, while carbon accounting smart contracts automate the calculation and update processes, notifying stakeholders of any changes.

4. SYSTEM TESTING AND IMPLEMENTATION

This section presents the results of the functions and events incorporated into our proposed solution. All smart contract functions were executed using the Remix IDE, and their results were verified. To improve security, we implemented modifiers within the smart contracts to restrict access to certain functions for unauthorized users. Any unauthorized access attempts trigger an error. Additionally, we integrated events within the smart contracts to ensure data provenance and enhance traceability. Each stakeholder and smart contract is assigned a unique Ethereum address, facilitating clear identification of entities. Table 1 presents the list of ethereum address of the stakeholders presented in this study.

Figure 3 highlights the successful registration of air cargo carrier and freight rail provider. The figure highlights the invoking of smart contract functions by the regulatory authority along with time stamp and block hash, which make the business transaction tamper-proof, transparent, and trust-worthy. Figure 4 presents successful execution of carbon accounting of scope 1, 2, and 3 emissions by the smart contract when the air cargo carrier invokes the

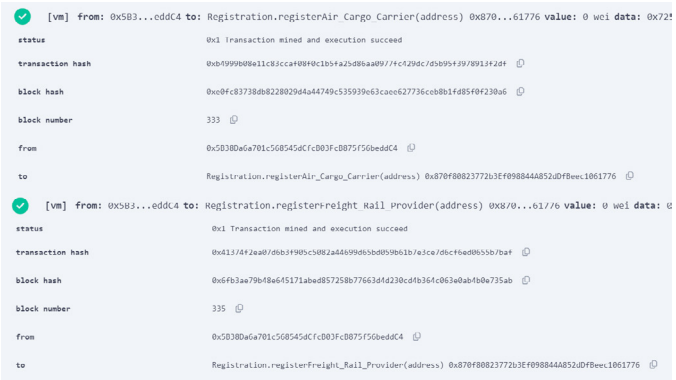

Fig. 3. Log showing successful registration of stakeholders

function *updateAirCO2Emissions*. The air cargo carrier updates related Scope 1, 2, and 3 emissions generated during the intermodal transportation processes and services. Based on the carbon accounting logic coded in the smart contract, the carbon accounting smart contract calculates the carbon emissions and notifies all the stakeholders about air cargo carrier's carbon emissions. Similarly, all stakeholders can invoke their respective functions and perform the carbon accounting of emissions in a decentralized, automated, and transparent manner.

4.1 Generalisation

The proposed blockchain-based system can be used to trace and account for carbon footprints within intermodal transportation supply chains, and has been developed, tested, and validated on the Ethereum platform. The system ensures the tracking of Scope 1, 2, and 3 emissions across a variety of transportation modes, allowing for a more accurate accounting of carbon emissions throughout an organization's supply chain. Transport operators and logistics providers can tailor the framework and smart contracts to suit their specific needs.

Fig. 4. Log showing successful execution of carbon accounting for air cargo carrier

Beyond transportation, this solution can be applied to a variety of industries to support carbon traceability and stakeholder accountability. It can be used for monitoring vessel emissions and optimizing fuel efficiency in maritime shipping, railroads can monitor energy consumption and electrification impacts, and public transportation and ride-sharing systems can integrate real-time carbon tracking. The framework can also be extended to industries such as construction, mining, and power generation to monitor emissions, calculate offsets, and comply with environmental regulations. We implemented smart contracts on the Ethereum blockchain, ensuring flexibility and scalability in carbon tracking. With slight modifications to emission accounting methodologies, the system can also accommodate other greenhouse gases and pollutants such as carbon monoxide and sulfur oxides, making it adaptable across various blockchain platforms. By offering a structured approach to carbon accounting, this framework provides a scalable and transparent solution for industries aiming to enhance sustainability, regulatory compliance, and emission reduction efforts.

5. CONCLUSION

This paper proposes a blockchain-based carbon accounting framework for tracking and accounting for Scope 1, 2, and 3 carbon emissions in intermodal transportation. By leveraging blockchain's inherent transparency, reliability, and security, our approach enhances the carbon accounting process across diverse modes of transportation, providing insights to address carbon accounting challenges. A sequence diagram is provided to illustrate how Ethereum-based smart contracts enable seamless tracking and accounting of carbon emissions across the intermodal network. The framework is flexible and can be implemented with minimal adjustments across different transportation systems; however, a few limitations remain. First, The scalability of the blockchain-based system presents a challenge due to high computational requirements, which lead to slower transaction speeds and increased storage needs. As the number of participants and transactions grows, the blockchain must process more data, which can cause delays and raise costs. Second, integrating IoT and AI for automated emissions verification is still in its early stages and requires further testing for large-scale deployment. Third, interoperability challenges exist when integrating blockchain with existing transportation management systems and regulatory platforms. Future research will explore large-scale deployment of this blockchain-based framework in intermodal transportation, incorporating IoT sensors for automated emissions tracking and AI-driven anomaly detection to prevent data manipulation. Moreover, developing decentralized verification mechanisms and incentive structures will further enhance reporting accuracy. Integrating this system with regulatory compliance platforms can streamline carbon auditing and policy enforcement.

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